

Research Paper

Clinical correlation of optic nerve head analysis performed by SD-OCT in children with pseudopapilledema and papilledema

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ABSTRACT

Purpose: To investigate some diagnostic indicators in optic nerve head (ONH) analysis of children with 'Pseudopapilledema' and 'Papilledema' by Spectral Domain Optical Coherence Tomography (SD-OCT).

Methods: Medical records of 52 children with optic disc swelling, who were followed up by the departments of pediatric neurology and ophthalmology between May 2018 and May 2019 were reviewed retrospectively. Cases were classified as group 1 (Papilledema secondary to Idiopathic Intracranial Hypertension) including 54 eyes of 27 cases, group 2 (Pseudopapilledema secondary to bilateral optic disc drusen) including 50 eyes of 25 cases and control group. Three-dimensional imaging of ONH, peripapillary retinal nerve fiber layer (RNLF) thickness, Bruch's membrane opening height in the nasal respectively temporal quadrant (BMH—N, BMH—T) levels were measured by SD-OCT and B-mode ocular ultrasonography data of all cases were evaluated.

Results: When RNFL levels were compared between groups, nasal RNFL levels were found to be significantly higher in group 1 cases compared to group 2 ($p < 0.001$). In 3D imaging of the ONH among group 1 patients, the mean height of the apex in disc swelling was significantly higher than that of group 2 ($p = 0.024$). The apex in disc swelling was mostly observed to be localized at the middle and had a diffuse swelling pattern in group 1; whereas, in group 2, it was localized mostly at the nasal quadrant. This variation was statistically significant ($p < 0.001$). When, Bruch's membrane opening height in the nasal respectively temporal quadrant (BMH—N, BMH—T) levels were compared, the measurements obtained were found to be significantly higher in group 1 ($p = 0.050$ and $p = 0.003$ respectively).

Conclusion: Nasal RNFL values of SD-OCT, Bruch's membrane opening height in the nasal respectively temporal quadrant (BMH—N, BMH—T) levels, the location of the apex in disc swelling obtained by 3D analysis of the ONH are found to be potential diagnostic parameters when combined with clinical findings. It is important that the nasal quadrant elevation, where the highest peak in 3D imaging was measured, was higher in the papilledema group. The elevation of the optic disc peak in 3D imaging can be used as a parameter to help clinicians distinguish between optic disc drusen (ODD) and papilledema.

1. Introduction

Although optic disc edema might occur with many different causes, bilateral optic disc edema caused by increased intracranial pressure is called 'Papilledema'. The basic fundoscopic findings of an established papilledema can be listed as blurring of the optic disc borders, swelling at the optic disc head, sometimes peripapillary hemorrhages with or without chorioretinal folds [1–3].

One of the most common pathologies in the differential diagnosis of papilledema is 'Pseudopapilledema caused by Optic Disc Drusen (ODD)'

and it is always a great challenge to differentiate these cases from early cases of papilledema.

Optic Disc Drusen was first described by Liebreich in 1868. It is usually bilateral in 75 % of cases and is more common in the Caucasian Race [4]. Abnormal axonal metabolism is considered to cause mitochondrial calcification in its pathophysiology [5,6]. However, calcium and amorphous extracellular components accumulate in congenitally prone optic discs to form drusen after axonal rupture [6]. Although the exact mechanism of drusen formation has not been demonstrated, problems such as a small scleral canal, congenital dysplastic disc,

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abnormal vascular structure, and impaired axoplasmic transport were reported as possible effective factors in its pathogenesis [6–11]. It may be difficult to detect it in early childhood and distinguish it from true papilledema because it is mostly embedded in the optic disc [12,13]. Although the incidence was found to be 3.4–4.9 % in clinical studies, it was reported to be 20.4 % in autopsies [14,15]. Optic disc drusen may compress the overlying retinal nerve fiber layer mechanically, and cause scotomas in the visual field such as nasal step, arcuate scotoma, blind spot enlargement, and concentric narrowing at different rates [7,14,16]. Although it is generally asymptomatic, there are studies reporting progressive visual field loss resulting from optic disc drusen in some patients [14,16].

Some of the other important conditions that can cause optic disc swelling other than papilledema including congenital optic disc swelling, high hypermetropia, disc tilting, myelinated nerve fibers, Hypertensive retinopathy, Central retinal vein occlusion, Anterior ischemic optic neuropathy, Diabetic papillopathy, Carotid-cavernous fistula, Inflammatory optic neuropathy [17].

During a neuro-ophthalmologic examination, tests such as confrontation or Humphrey visual field testing, spectral domain optical coherence tomography for retinal nerve fiber layer measurement (SD-OCT), fundus fluorescein angiography (FFA), B-scan orbital ultrasonography, contrast sensitivity testing (KDT), photostress test, electroretinography (ERG), visual evoked potentials (VEP), MRI + Gd contrast agent, and standard funduscopy can be utilized to assist in making a diagnosis [18–20].

The primary approach to treating papilledema medically involves a combination of medication therapy, weight loss, and dietary modifications [21,23].

Carbonic anhydrase inhibitors, a group that reduces cerebrospinal fluid (CSF) production, are primarily used in cases of severe visual loss and chronic papilledema. For patients who cannot tolerate carbonic anhydrase inhibitors, furosemide (adult dose ranging from 40 to 80 mg daily) may be administered. Furosemide is believed to reduce CSF volume by decreasing sodium transport from the choroid plexus. The adult dose range for acetazolamide is 1000 mg to 2000 mg daily. It is contraindicated in chronic obstructive pulmonary diseases, liver or kidney diseases, pregnancy, and sulfonamide allergy. Potassium supplementation and increased fluid intake are recommended when taking acetazolamide [17,22,23].

Distinguishing papilledema from pseudopapilledema is important because the former is usually not harmful, whereas the latter may be quite harmful.

The purpose of the present study was to investigate the diagnostic effectivity of some markers of Optic Nerve Head (ONH) analysis in Optical Coherence Tomography (SD-OCT) in pediatric cases with ODD and true papilledema to facilitate the clinical diagnosis.

2. Materials and methods

The study had a retrospective design. The approval of Ufuk University Faculty of Medicine Dr. Rudvan Ege Research Hospital Ethics Committee was received for the study (Project application date/number: 20,190,703/1). In line with the World Medical Association Declaration of Helsinki, written informed consent was obtained from at least 1 parent of all cases included in the study [24].

Medical records of 52 patients diagnosed with childhood true and pseudopapilledema by ODD, followed up by the Ophthalmology and Child Neurology Department between May 2018 and May 2019 were reviewed retrospectively. These children were admitted to the hospital with complaints of headache.

The cases were classified as group 1 (true Papilledema cases secondary to Idiopathic Intracranial Hypertension-IIH), group 2 (Pseudopapilledema cases secondary to bilateral ODD), and group 3 (control group). control group was designed from those who were under the age of 18 and were presented to Ophthalmology Clinic for various reasons

except optic nerve, vitreoretinal, or cerebrovascular diseases. The data of 54 eyes of 27 cases diagnosed with true papilledema (by IHH) were analyzed as group 1, the data of 50 eyes of 25 cases diagnosed with pseudo-papilledema (by bilateral ODD) were analyzed as group 2, and the data of 52 eyes of 26 healthy cases were analyzed as group 3.

2.1. Inclusion criteria of the study

- Being under 18 years of age
- Spherical equivalent between -1.25 and $+1.25$ diopters in refractive errors
- Not having undergone ocular surgery
- Having corrected intraocular pressure as <20 mmHg
- Having the best corrected visual acuity as 20/20
- Not having media opacity that might affect SD-OCT images (advanced cataract, corneal opacity, vitreous opacity, etc.)
- Being compatible with SD-OCT imaging and B-mode ophthalmic ultrasonography measurement
- Having a shooting quality in SD-OCT imaging of 6/10 and above
- Not having any obstacles to neuroradiological imaging
- Having the diagnoses of true papilledema or pseudopapilledema by common consensus after neurological and ophthalmological evaluation
- The basic ophthalmological examination records, including visual acuity examinations detected with Snellen Chart, biomicroscopy, intraocular pressure measurements with Goldmann Applanation Tonometer, and dilated fundus examinations, were examined for all true or pseudopapilledema cases.

Detailed ophthalmological examinations and all SD-OCT measurements of the patients were performed by the same ophthalmologist (Dr II) at Ufuk University Faculty of Medicine Ophthalmology Clinic between 9.00 and 11.00 am.

The Cirrus HD-OCT (Carl Zeiss Meditec, Model 400, Dublin, USA, Software 8.1.0.117) Optical Coherence Tomography (OCT) Device was used for 3D imaging analysis of the Optic Nerve Head, peripapillary retinal nerve fiber layer thickness, central macular thickness, submacular ganglion cell layer analysis, *en face* images of optic nerves. The measurements were made in Optic Disc Cube 200×200 mode. RNFL scans were recorded from a 3.4 mm circular area, keeping the optic nerve in the center. The measurements were automatically made by the device in 4 quadrants as well as the "average" of RNFL thickness over the 4 quadrants as a fifth measurement. Each quadrant and average RNFL thickness values were analyzed separately in statistical analyses.

The height between the Bruch's Membrane Opening Site in the *en face* image of the retinal nerve fiber and the peak of the highest nerve fiber quadrant was measured manually in the 3D images of the optic disc head.

B-mode Ophthalmic Ultrasonography (Sonomed EZ SCAN AB5500⁺) was used for ultrasonic evaluation in all patients, first with 80 dB and then with 60 dB gain. The results were evaluated by two independent observers, and the correlation between them was evaluated in statistical terms.

The information in the En Face Analysis of the 200×200 Optical Disc Cube in the Cirrus HD-OCT Device was retrieved. A perpendicular line was drawn from the opening of the nasal part of Bruch's Membrane to the highest point of the optic disc nasal and the distance was measured. Similarly, a perpendicular line was drawn from the opening of the temporal part of Bruch's Membrane to the highest point of the optic disc temporal and the distance was measured (Fig. 1).

Information in Cirrus HD-OCT (200×200 Optical Disk Cube 3D Imaging Analysis) was retrieved (Fig. 2). The width of the cube on the screen in 3D imaging was between 1 and 200 μm , the height was between 1 and 175 μm , and A-Scan was between 1 and 1024 μm . The perpendicular projection of the highest point of the disc to the internal limiting membrane was calculated manually in 3D imaging. Which of the inferior, superior, temporal, and nasal quadrants the highest point was located was identified by moving the number bars at the corners of

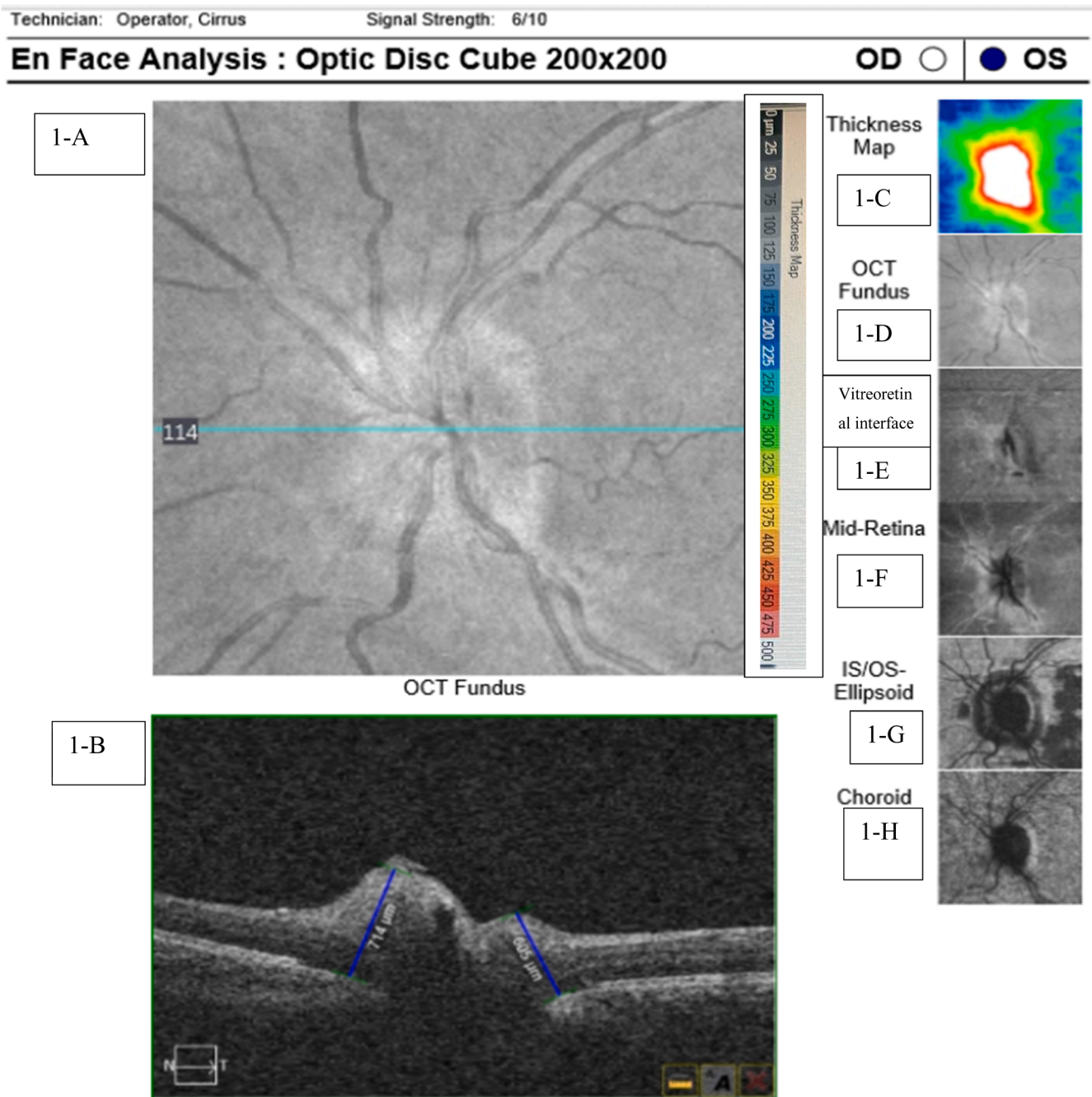


Fig. 1. An example of the OCT image of En-Face Analysis of a patient with papilledema which indicates greater nasal quadrant swelling than temporal quadrant (1-A, 1-B, 1-C, 1-D). Vitreoretinal interface, mid-retina, IS/OS Ellipsoid and Choroid image (1-E, 1-F, 1-G, 1-H, respectively).

the cube on the 3D screen (Fig. 3).

2.2. Statistical analysis

Whether the distribution of the continuous numerical variables was close to normal was examined with the Kolmogorov-Smirnov Test and whether the assumption of homogeneity of the variances was met was investigated with the Levine Test. Descriptive statistics were expressed as mean $\pm$ standard deviation or median (interquartile distribution width) for continuous numerical variables, and categorical variables were shown as number of cases and (%).

As a result of the Goodness of Fit Tests, the significance of the differences between the groups in terms of continuous numerical variables for which parametric test statistics assumptions were met was evaluated

by the One-Way Analysis of Variance (One-Way ANOVA). The significance of the differences between the groups for continuous numerical variables for which parametric test statistics assumptions were not met was evaluated with the Mann-Whitney Test when the number of independent groups was two. The significance of the differences between more than two independent groups was examined with the U-Test and the Kruskal-Wallis Test. If the results of the One-Way Analysis of Variance or Kruskal-Wallis Test Statistics were found to be significant, the group(s) that caused the difference were determined by using the Post-Hoc Tukey HSD or Dunn-Bonferroni Test.

Whether there were statistically significant differences between the nasal and temporal regions in terms of height (H) measurements of the Bruch's Membrane opening sites at the optic disc temporal and nasal (respectively) was evaluated with the Wilcoxon Sign Test. The

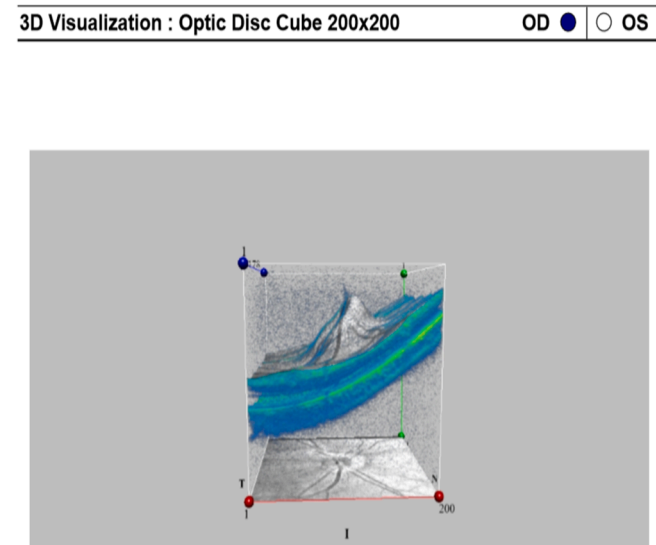


Fig. 2. An example of the OCT image of 3D optic disc imaging of a patient with papilledema which indicates greater nasal quadrant swelling than temporal quadrant.

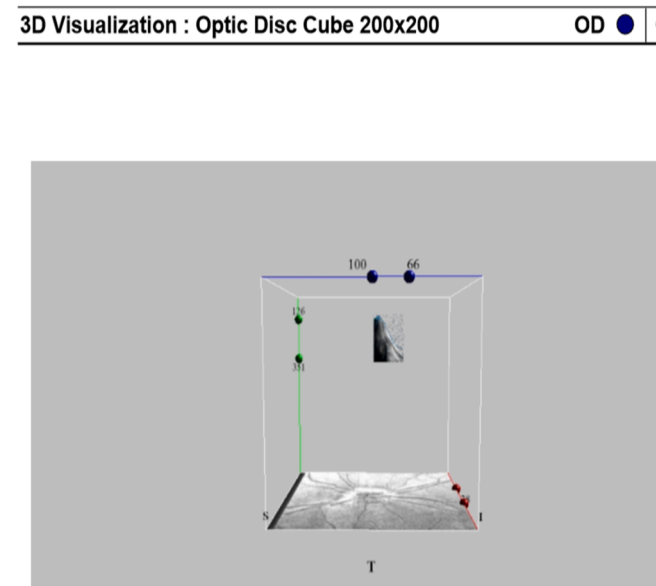


Fig. 3. An example of an OCT image of a patient with papilledema, where the highest point in 3D imaging was determined in the nasal quadrant.

significance of the differences in the distribution of quadrant localizations where the highest point in the 3D imaging of the optic discs was found was evaluated with Cochran's Q-Test. If Cochran's Q-Test results were significant, the region(s) causing the difference were determined by using the McNemar Test.

The analysis of the data was carried out with the IBM SPSS Statistics 17.0 package program (IBM Corporation, Armonk, NY, USA). The results were considered statistically significant at $p < 0.05$.

3. Results

Mean RNFL levels between the groups were significantly different ($p < 0.001$). Mean RNFL levels in group 1 and group 2 were higher than that of the control group ($p = 0.038$ and $p < 0.001$, retrospectively) (Table 1).

A statistically significant difference was detected between the groups

Table 1

RNFL levels of the cases according to groups.

	Control ($n = 26$)	ODD ($n = 25$)	Papilledema ($n = 27$)	p-value †
Nasal	68.2 (5.38) ^a [μ]	63.0 (22.25) ^b [μ]	98.0 (75.50) ^{a,b} [μ]	<0.001
Superior	123.5 (16.75) [μ]	126.0 (29.00) [μ]	147.5 (166.50) [μ]	0.006
Temporal	67.0 (13.00) ^{a,c} [μ]	74.5 (8.25) ^c [μ]	74.0 (31.50) [μ]	<0.001
Inferior	113.0 (14.88) ^{a,c} [μ]	129.5 (27.25) ^c [μ]	138.5 (145.50) [μ]	<0.001
Mean	92.0 (6.75) ^{a,c} [μ]	97.5 (13.00) ^c [μ]	116.5 (109.00) [μ]	<0.001

Data are shown as median (interquartile range), † Kruskal Wallis Test, a: Statistically significant, [μ]: micron difference between the Control Group and Papilledema ($p < 0.01$), b: Statistically significant difference between the ODD Group and Papilledema ($p < 0.001$), c: Statistically significant difference between the Control Group and ODD ($p < 0.05$).

in terms of nasal RNFL levels ($p < 0.001$), and the reason for this difference was the higher nasal RNFL level of group 1 when compared group 2 and group 3 ($p < 0.001$ and $p < 0.001$) (Table 1).

When compared to group 2, the height in the quadrant where the highest peak in 3D imaging was measured in group 1 was higher at a statistically significant level ($p = 0.024$). Both, Bruch's membrane opening height in the nasal respectively temporal quadrant (BMH—N, BMH—T) levels of group 1 were higher when compared to group 2 at statistically significant levels ($p = 0.050$ and $p = 0.003$) (Fig. 4) (Table 2).

The frequency of the peak location being diffuse and in the middle in 3D imaging in group 1 was higher at a statistically significant level ($p < 0.001$). The frequency of the nasal location of the peak in 3D imaging was higher in group 2 at a statistically significant level ($p < 0.001$) (Table 3).

The area under the ROC Curve of nasal RNFL measurements was statistically significant in distinguishing group 1 and 2 [AUC: 0.813, 95 % Confidence Interval: 0.694–0.931 and $p < 0.001$] The area under the ROC Curve of the mean RNFL measurements was statistically significant in distinguishing group 1 and 2 [AUC: 0.708, 95 % Confidence Interval: 0.563–0.854 and $p = 0.010$] (Fig. 5) (Table 4).

The area under the ROC Curve in peak height measurements in 3D imaging was statistically significant in distinguishing group 1 and 2 [AUC: 0.683, 95 % Confidence Interval: 0.535–0.831 and $p = 0.024$]. The area under the ROC Curve in Bruch's membrane opening height in the nasal (BMH—N) measurements was statistically significant in distinguishing group 1 and 2 [AUC: 0.659, 95 % Confidence Interval:

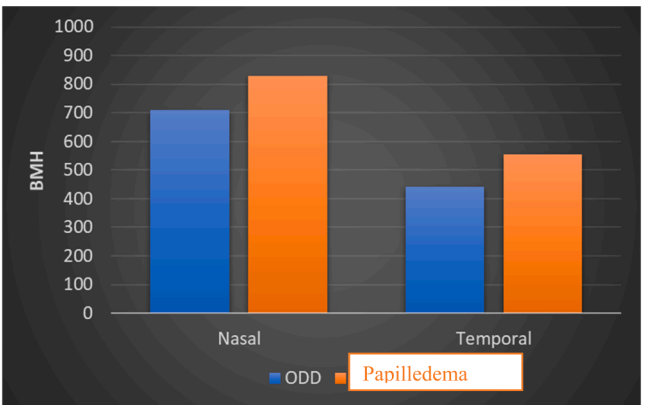


Fig. 4. Temporal and nasal BMH(Bruch's membrane opening height in the nasal respectively temporal quadrant) levels of the ODD and papilledema Groups.

Table 2
OCT findings of the cases according to groups.

	Control (n = 26)	ODD (n = 25)	Papilledema (n = 27)	p-value
Heights of the peak point in 3D imaging in the quadrant where it was measured the highest	–	222.5 (81.50) [μ]	266.5 (155.00) [μ]	0.024 ¶
BMH- Nasal **	–	710.0 (164.25) [μ]	830.0 (373.00) [μ]	0.050 ¶
BMH- Temporal **	–	441.5 (157.25) [μ]	553.5 (493.00) [μ]	0.003 ¶

*Descriptive statistics are expressed as mean ± standard deviation, ** Bruch's membrane opening height in the nasal respectively temporal quadrant (BMH–N, BMH–T) levels -The data are shown as median-interquartile range width, † One-Way ANOVA, ‡ Kruskal Wallis Test, ¶ Mann Whitney U Test.
-The data are shown as median-interquartile range width, † One-Way ANOVA, ‡ Kruskal Wallis Test, ¶ Mann Whitney U Test.
[μ]:micron.

Table 3
Distribution of the cases according to the location of the highest point in 3D imaging.

	ODD (n = 25)	Papilledema (n = 27)	p-value
Diffuse in the middle	0 (0.0 %)	10 (37.0 %)	<0.001 †
Nasal	25 (100.0 %)	17 (63.0 %)	<0.001 †
Inferior	11 (44.0 %)	6 (22.2 %)	0.169 ‡
Superior	18 (72.0 %)	14 (51.9 %)	0.227 ‡
Temporal	2 (8.0 %)	4 (14.8 %)	0.670 ‡
Inferior	1 (4.0 %)	2 (7.4 %)	>0.999 ‡
Superior	1 (4.0 %)	2 (7.4 %)	>0.999 ‡

† Fisher's Exact Probability Test.
‡ Chi-Square Test with continuity correction.

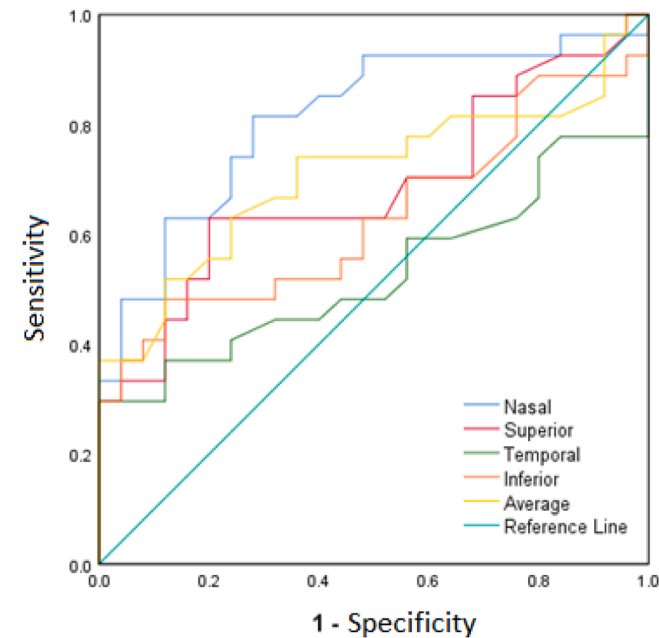


Fig. 5. The area under the ROC curve for RNFL measurements in distinguishing between papilledema and ODD(optic disc drusen) groups.

0.501–0.816 and $p = 0.050$]. The area under the ROC Curve in Bruch's membrane opening height in temporal quadrant (BMH-T) measurements was statistically significant in distinguishing group 1 and 2 [AUC:

0.743, 95 % Confidence Interval: 0.606–0.880 and $p = 0.003$] (Table 5).

4. Discussion

In our study nasal RNFL values of SD-OCT, Bruch's membrane opening height in the nasal respectively temporal quadrant (BMH–N and BMH–T) levels, the location of the apex in disc swelling obtained by 3D analysis of the ONH are found to be potential diagnostic parameters when combined with clinical findings. In addition to these findings the nasal quadrant elevation, where the highest peak in 3D imaging was measured, was higher in the papilledema group. The elevation of the optic disc peak in 3D imaging can be used as a parameter to help clinicians distinguish between optic disc drusen (ODD) and papilledema.

Significantly higher nasal RNFL levels were detected in patients with true papilledema when compared to patients with ODD and normal subjects. Nasal RNFL levels were statistically similar between patients with ODD and normal subjects. In their study, Bassi ST and Mohana KP also reported similar results among these groups in terms of nasal RNFL levels [25].

When Areas Under the ROC Curve were evaluated in statistical terms, the average RNFL and all RNFL quadrant values had highest diagnostic accuracy rates in distinguishing between the patients with papilledema and control group, and the highest accuracy rate was the nasal quadrant RNFL values. Based on these results, we propose that nasal RNFL level detected by SD-OCT is an effective OCT parameter in distinguishing papilledema from ODD.

In addition, it was found that the superior RNFL levels in papilledema group was significantly elevated than the control group, and this difference was not observed among other groups. Superior RNFL levels were similarly reported to be significantly higher in papilledema group when compared to pseudopapilledema patients and this difference was significantly higher than that of the control group in the study of Bassi ST and Mohana KP [25]. In another study that was conducted by Masoud Aghsaei Fard et al., superior RNFL levels were found to be significantly elevated in the papilledema group than in both the pseudopapilledema and control groups, and no significant differences were detected in the pseudopapilledema group when compared to the control group [26].

In our study, the mean RNFL levels of the ODD and papilledema groups were found to be significantly elevated than the control group, however, the mean RNFL levels were similar among patients with ODD and papilledema. In their studies, Bassi ST, Mohana KP [25] and Masoud Aghsaei Fard et al. [26] detected that the mean RNFL levels were significantly higher in papilledema group compared to the control and pseudopapilledema groups although a significant increase was not detected in the pseudopapilledema group compared to the control group [26]. The results of our study does not completely support this finding since a difference in mean RNFL was not documented between ODD and papilledema patients. This may probably be attributed to the early stages of nerve fiber edema in most of our cases involved in papilledema group. Cases at early stages of papilledema always have the most diagnostic challenge in differential diagnosis from ODD and our study was mainly conducted among this challenging group which may be one of the STRONG aspects of our study.

RNFL that is measured with the SD-OCT Device and Optic Nerve Head *en-face* analyses are used frequently in Optic Nerve Head evaluations. In our study, morphometric analysis of the Optic Nerve Head was performed with the 3D imaging mode on the SD-OCT Device, which is a previously unstudied parameter in the literature for the differential diagnosis of optic nerve diseases. The perpendicular projection of the highest point of the optic disc to the internal limiting membrane was calculated manually with 3D imaging mode in this analysis. The highest point was found in the inferior, superior, temporal, and nasal quadrants by using the manual calculation method by moving the number of indicators at the corners of the cube on the 3D viewing screen. The measurement algorithm was defined by us and used for the first time in the literature because similar studies have not been conducted before.

Table 4

The area under the ROC curve, 95 % Confidence Intervals, best cut-off points, and diagnostic performance indicators for RNFL measurements to distinguish between groups.

	Nasal	Superior	Temporal	Inferior	Average
Control vs Papilledema					
AUC	0.830	0.738	0.739	0.830	0.827
95 % CI	0.703–0.957	0.594–0.882	0.603–0.876	0.714–0.946	0.704–0.950
p-value	<0.001	0.003	0.003	<0.001	<0.001
Cut-off	>78	>140.25	>73.25	>122.75	>104.25
Sensitivity	74.1 %	63.0 %	55.6 %	77.8 %	66.7 %
Selectivity	96.2 %	100.0 %	92.3 %	80.8 %	100.0 %
PEV	95.2 %	100.0 %	88.2 %	80.8 %	100.0 %
NEV	78.1 %	72.2 %	66.7 %	77.8 %	74.3 %
Accuracy	84.9 %	81.2 %	73.6 %	79.2 %	83.1 %
Control vs ODD					
AUC	0.588	0.591	0.801	0.802	0.747
95 % CI	0.417–0.759	0.427–0.754	0.681–0.920	0.680–0.924	0.611–0.883
p-value	0.283	0.266	<0.001	<0.001	0.002
Cut-off	–	–	>73.50	>122.75	>93.75
Sensitivity	–	–	56.0 %	76.0 %	72.0 %
Selectivity	–	–	92.3 %	80.8 %	69.2 %
PEV	–	–	87.5 %	79.2 %	69.2 %
NEV	–	–	68.6 %	77.8 %	72.0 %
Accuracy	–	–	74.6 %	78.5 %	70.6 %
ODD vs Papilledema					
AUC	0.813	0.682	0.528	0.632	0.708
95 % CI	0.694–0.931	0.535–0.830	0.364–0.692	0.477–0.786	0.563–0.854
p-value	<0.001	0.024	0.728	0.103	0.010
Cut-off	>71.5	>142.25	–	–	>114.75
Sensitivity	81.5 %	63.0 %	–	–	51.9 %
Selectivity	72.0 %	80.0 %	–	–	88.0 %
PEV	75.9 %	77.3 %	–	–	82.4 %
NEV	78.3 %	66.7 %	–	–	62.9 %
Accuracy	76.9 %	71.2 %	–	–	69.2 %

AUC: Area Under the Curve, CI: Confidence Interval, PEV: Positive Estimated Value, NEV: Negative Estimated Value.

Table 5

Area under the ROC curve, 95 % confidence intervals, best cut-off points, and diagnostic performance indicators in peak height and BMH measurements on 3D imaging to distinguish ODD and papilledema groups.

	Peak height in 3D Imaging	BMH nasal*	BMH temporal*
AUC	0.683	0.659	0.743
95 % CI	0.535–0.831	0.501–0.816	0.606–0.880
p-value	0.024	0.050	0.003
Cut-off	>291	>855.5	>543.75
Sensitivity	48.1 %	44.4 %	59.3 %
Selectivity	92.0 %	100.0 %	88.0 %
PEV	86.7 %	100.0 %	84.2 %
NEV	62.2 %	62.5 %	66.7 %
Accuracy	69.2 %	71.2 %	73.1 %

AUC: Area Under the Curve, CI: Confidence Interval, PEV: Positive Estimated Value, NEV: Negative Estimated Value.

* Bruch's membrane opening height in the nasal respectively temporal quadrant (BMH–N, BMH–T) levels.

When this method is used for measurements, the quadrant height at which the highest peak in 3D imaging was measured was found to be significantly elevated in cases with papilledema than in cases with ODD. Although the mean RNFL levels were similar in papilledema and ODD cases in our study, it is important that the quadrant elevation where the highest peak in 3D imaging was measured higher in papilledema group. The elevation of the optic disc peak in 3D imaging can be used as a parameter to give clinicians an idea of the distinction between ODD and papilledema. The effectiveness of this newly described parameter needs to be investigated further in larger series and controlled studies.

3D imaging of the Optic Disc Head might be an efficient novel technique in differentiating ODD and papilledema because it is a practical and non-invasive method. In our study, it was found in 3D imaging that all quadrants were swollen at the head of the optic disc, the peak was mostly at the middle level of all quadrants, and a “diffuse height

peak image” was detected in the majority of the cases in the PTC group. A peak image with the peak leaning towards the nasal quadrant was found in the ODD group because the peak projection fell more on the nasal quadrant and each quadrant did not rise diffusely. On the bulging nasal quadrant, the RNFL value was detected as thinner than the other quadrants. The reason for thin RNFL in the nasal quadrant in ODD cases can be explained in this way. It is considered that this is why cases with high nasal quadrant RNFL values are often diagnosed with papilledema. A similar hypothesis was presented in the literature by Kyoung Min Lee, however, this association could not be elucidated in this study because the 3D analysis of the Optic Nerve Head and its peak image leaning on the nasal quadrant were not revealed [27]. We think that our study will make a significant contribution to the literature data in this respect.

The distances of Bruch's Membrane opening sites to the temporal and nasal quadrants of the Optic Nerve Head are parameters that can be measured by SD-OCT. However, these were previously used to evaluate cases with optic disc swelling [21]. In the study conducted by Pardon LP et al., based on Bruch's Membrane opening points, the minimum rim width was found to be significantly greater in the papilledema group than in the pseudopapilledema and control groups and was significantly greater in the pseudopapilledema group compared to the control group [28].

In our study, Bruch's membrane opening height in the nasal respectively temporal quadrant (BMH–N, BMH–T) levels of papilledema cases were higher than cases with ODD at statistically significant levels. When the distances of the Bruch's Membrane opening points to the temporal and nasal quadrants of the Optic Nerve Head were measured, the results were significantly higher in papilledema group, which can be used as another parameter that can be used in differential diagnosis.

Controlled studies involving larger series are needed to establish more diagnostic OCT parameters in the differential diagnosis of papilledema in cases presenting with optic disc swelling. Future studies may even involve artificial intelligence in this topic. We believe that the

assistance of modern technology will be soon guiding for the early diagnosis and to initiate the effective treatment in these challenging cases.

The limitations of the present study were the small number of patients, the high number of volunteers in the group participating in the study in favor of females in the gender distribution. staging the cases with papilledema was not performed and most papilledema cases were those in early stages of edema which had a diagnostic challenge in differential diagnosis of odd.

5. Conclusion

Optic disc swelling may occur because of many etiologies, including true or pseudopapilledema. Determining effective and accurate differential diagnostic criteria in these cases will prevent unnecessary examinations and procedures during the diagnosis. When combined with clinical findings, nasal RNFL values that are measured by SD-OCT, Bruch's membrane opening height in the temporal quadrant (BMH-T) levels and the location of the bulge peak in 3D analysis of the Optic Nerve Head can be used as guiding parameters in differential diagnosis.

We think that the localization of the highest point (peak) in the optic disc bulging detected with this 3D analysis imaging and, Bruch's membrane opening height in the temporal quadrant (BMH-T) levels may play extremely important roles in the future, in distinguishing these two groups.

The results from this paper have been presented to the Ankara Ufuk University as a thesis with interim findings. The thesis is available on the institution's repository.

Raw data were generated at Ankara Ufuk University, Derived data supporting the findings of this study are available from the corresponding author on request.

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Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.: Pınar alıyaylık ozer, İlter iritaş, Çiğdem Genç Sel.

CRediT authorship contribution statement

İlter Iritas: Writing – review & editing, Writing – original draft, Resources, Methodology, Investigation, Formal analysis, Data curation.
Pınar A. Ozer: Writing – review & editing, Supervision, Data curation.
Çiğdem Genç Sel: Methodology, Investigation, Data curation.

Declaration of competing interest

The authors have no conflicts of interest to declare.

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